

FINANCE · UNIT 2

Valuation & Time Value of Money

A complete, deeply explained guide to how money changes value over time, how to calculate present and future value, opportunity cost of capital, and how to value any asset — built for absolute beginners.

Sections Covered: 2.1 Time Value of Money · 2.2 Present Value (PV) · 2.3 Future Value (FV) · 2.4 Opportunity Cost of Capital · 2.5 Valuation of Assets · 2.6 Risk & Return

Includes diagrams, worked examples, glossary, and a 12-question practice quiz

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Valuation & Time Value of Money

Unit 2 — Complete Study Guide

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1

Time Value of Money

The **Time Value of Money (TVM)** is the single most fundamental concept in all of finance. Every calculation in this unit — present value, future value, valuation of assets, opportunity cost — is built on this one idea: **a sum of money available today is worth more than the identical sum available in the future.**

This is not an opinion or a cultural convention. It is a mathematical and economic reality that applies universally — in Tashkent, Tokyo, or New York. Once you truly understand the time value of money, you will never look at financial decisions the same way again.

The Three Reasons Money Has a Time Value

There are exactly three reasons why money today is worth more than money tomorrow. All three are real and all three operate simultaneously in every financial market.

- **Reason 1: Investment Opportunity:** Money you hold today can be put to work immediately. You can deposit it in a bank, buy assets, start a business, or invest in stocks. Each of these generates a return. A sum that sits idle in the future produces nothing today. This is the most important reason — money has productive potential, and time is required to realize it.
- **Reason 2: Inflation:** In virtually every economy, prices rise over time. In Uzbekistan, inflation averaged 8–12% annually in recent years. This means 1,000,000 UZS today buys more goods and services than 1,000,000 UZS will in five years. The purchasing power of money erodes continuously. Holding uninvested cash is therefore not neutral — it is a guaranteed loss in real terms.
- **Reason 3: Risk and Uncertainty:** The future is uncertain. A promise of money in the future carries the risk that it may never arrive: the borrower may default, the investment may fail, political or economic events may intervene. Money held today is certain. Money promised in the future is not. Rational people require compensation for this uncertainty — which is why future money is discounted.

■ The Core Principle of TVM

Given a choice between receiving 1,000,000 UZS today or 1,000,000 UZS one year from now, a rational person always chooses today. Not out of impatience or greed — but because today's money can be invested to earn returns, its purchasing power is higher, and it carries no collection risk. This preference for present money over future money is called the **time preference for money**, and it is the foundation of everything in this unit.

A Concrete Illustration

Let's make this very concrete. Suppose you lend a friend 2,000,000 UZS today. He promises to return it in 3 years. What is the "fair" repayment?

If he returns exactly 2,000,000 UZS, you have actually lost money. Here is why:

- You gave up the ability to invest that money for 3 years. At a conservative 12% per year bank deposit, 2,000,000 UZS would have grown to $2,000,000 \times (1.12)^3 = 2,809,856$ UZS.
- Inflation at 8% per year means 2,000,000 UZS in 3 years buys what approximately 1,587,600 UZS buys today.
- You took on the risk that your friend might not pay at all.

The fair repayment is approximately 2,809,856 UZS — not 2,000,000. The extra 809,856 UZS is the compensation for time, inflation, and risk. **This is exactly what interest is** — not an arbitrary fee, but the mathematically correct price of borrowing money across time.

— ■ WORKED EXAMPLE — WORKED EXAMPLE — TVM in a Real Uzbekistan Context

Scenario: You have 10,000,000 UZS in cash under your mattress for 2 years.

Bank deposit rate: 14% per year. Inflation: 9% per year.

After 2 years, your cash is still 10,000,000 UZS nominally.

BUT: If you had deposited it, you would have: $10,000,000 \times (1.14)^2 = 12,996,000$ UZS.

AND: Due to inflation, your 10,000,000 UZS only buys what 8,416,800 UZS bought 2 years ago.

Net result of holding cash: You "lost" approximately 4,579,200 UZS in opportunity terms.

Lesson: There is no such thing as "safe" cash under a mattress. Inaction is a decision with a cost.

Why This Matters for Every Financial Decision

Once you understand TVM, you realize that **every financial decision involves comparing money at different points in time**. This is true whether you are:

- Deciding whether to buy a car now or save for it
- Evaluating whether a business investment is worthwhile
- Choosing between two job offers with different pay structures
- Deciding when to buy a house
- Comparing investment options with different time horizons

In every case, to compare fairly, you must convert all cash flows to the **same point in time**. That is what Present Value and Future Value calculations allow us to do.

2.2 Present Value (PV)

Present Value (PV) is the current value of a future sum of money, given a specific rate of return (the discount rate). In other words, it answers the question: **"How much is a payment I will receive in the future worth to me right now?"**

PV is one of the most powerful tools in all of finance because it allows us to compare money received at completely different points in time on a fair, equal basis. Without PV, we cannot evaluate investments, value businesses, price bonds, or make rational borrowing decisions.

The Present Value Formula

The formula for calculating Present Value is:

$$PV = FV \div (1 + r)^n$$

Symbol	Name	What It Means	Example
PV	Present Value	The value today — what we want to find	?
FV	Future Value	The cash amount to be received in the future	10,000,000 UZS
r	Discount Rate	The rate of return per period (as a decimal)	0.12 (= 12%)
n	Number of Periods	How many years (or periods) until payment	3
$(1+r)^n$	Discount Factor	The multiplier that converts future to present	1.404 for r=12%, n=3

Step-by-Step: How to Calculate PV

Let's work through a complete calculation step by step, so you can apply it to any situation.

— ■ WORKED EXAMPLE — WORKED EXAMPLE — Full PV Calculation

Problem: You will receive 15,000,000 UZS in 4 years. You can earn 10% per year on alternative investments. What is 15,000,000 UZS worth to you TODAY?

Step 1: Identify the inputs.

$$FV = 15,000,000 \quad r = 0.10 \quad n = 4$$

Step 2: Calculate the discount factor.

$$(1 + r)^n = (1.10)^4 = 1.10 \times 1.10 \times 1.10 \times 1.10 = 1.4641$$

Step 3: Divide FV by the discount factor.

$$PV = 15,000,000 \div 1.4641 = 10,245,181 \text{ UZS}$$

Interpretation: Receiving 15,000,000 UZS in 4 years is equivalent to receiving 10,245,181 UZS today, assuming you can earn 10% per year. If someone wants to sell you a right to receive 15M in 4 years, you should pay NO MORE than 10,245,181 UZS for it.

How the Discount Rate Affects PV

The discount rate is the most powerful variable in the PV formula. Even small changes in the rate produce large differences in present value, especially over long time horizons. The higher the rate, the less future money is worth today.

Discount Rate	PV of 10M UZS in 1 year	PV of 10M UZS in 5 years	PV of 10M UZS in 10 years
5%	9,523,810 UZS	7,835,262 UZS	6,139,133 UZS
10%	9,090,909 UZS	6,209,213 UZS	3,855,433 UZS
15%	8,695,652 UZS	4,971,767 UZS	2,471,847 UZS
20%	8,333,333 UZS	4,018,776 UZS	1,615,055 UZS
25%	8,000,000 UZS	3,276,800 UZS	1,073,742 UZS

Notice: at 25% discount rate over 10 years, 10,000,000 UZS in the future is worth only 1,073,742 UZS today — about 11 cents on the dollar. This illustrates why high-risk investments (which require high discount rates) place much less value on distant future cash flows.

What Is the Right Discount Rate to Use?

Choosing the correct discount rate is one of the most important and debated questions in finance. The discount rate should represent the **opportunity cost of capital** — the return you could earn on an alternative investment of similar risk.

- For a very safe investment: use a low rate (e.g., 10–14% in Uzbekistan, matching bank deposits)
- For a medium-risk investment (real estate, established business): use 15–20%

- For a high-risk investment (startup, crypto): use 25–40% or higher
- For valuing corporate projects: use the company's cost of capital (covered in Unit 3)

■ Key Insight on Discount Rates

If you use too LOW a discount rate, you will overvalue investments and overpay. If you use too HIGH a rate, you will undervalue them and miss opportunities. The art of finance is choosing the right rate that accurately reflects the risk of the cash flows being valued. This is why financial analysts spend enormous effort debating and calculating discount rates.

2. Future Value (FV)

Future Value (FV) is the mirror image of Present Value. Where PV asks "what is future money worth today?", FV asks **"what will today's money be worth in the future?"** Future value is the mathematical engine behind compound interest — and understanding it is the key to building wealth over time.

The Future Value Formula

$$FV = PV \times (1 + r)^n$$

— ■ WORKED EXAMPLE — WORKED EXAMPLE — Full FV Calculation

Problem: You invest 5,000,000 UZS today at 12% per year. How much will you have in 5 years?

Step 1: Identify inputs.

$$PV = 5,000,000 \quad r = 0.12 \quad n = 5$$

Step 2: Calculate $(1 + r)^n$.

$$(1.12)^5 = 1.12 \times 1.12 \times 1.12 \times 1.12 \times 1.12 = 1.7623$$

Step 3: Multiply PV by the growth factor.

$$FV = 5,000,000 \times 1.7623 = 8,811,709 \text{ UZS}$$

Your 5,000,000 UZS grows to 8,811,709 UZS in 5 years. The extra 3,811,709 UZS is the return on your investment — interest on interest.

Simple Interest vs. Compound Interest

Before going further, it is critical to understand the difference between simple and compound interest. They produce dramatically different outcomes, especially over long periods — and this difference is at the heart of why starting to invest early matters so much.

Simple Interest

With simple interest, you earn interest **ONLY** on the original amount (principal). No matter how long you invest, the interest earned each period is the same fixed amount.

$$FV (\text{Simple}) = PV \times (1 + r \times n)$$

Example: 1,000,000 UZS at 10% simple interest for 5 years.

- Year 1: $1,000,000 + 100,000 = 1,100,000$ UZS
- Year 2: $1,100,000 + 100,000 = 1,200,000$ UZS
- Year 5: $1,000,000 \times (1 + 0.10 \times 5) = 1,500,000$ UZS

Growth is **LINEAR** — the same 100,000 UZS every year.

Compound Interest

With compound interest, you earn interest on your principal AND on all previously earned interest. Your interest earns interest. This creates exponential (accelerating) growth.

$$\text{FV (Compound)} = \text{PV} \times (1 + r)^n$$

Example: 1,000,000 UZS at 10% compound interest for 5 years.

- Year 1: $1,000,000 \times 1.10 = 1,100,000$ UZS (interest: 100,000)
- Year 2: $1,100,000 \times 1.10 = 1,210,000$ UZS (interest: 110,000)
- Year 3: $1,210,000 \times 1.10 = 1,331,000$ UZS (interest: 121,000)
- Year 5: $1,000,000 \times (1.10)^5 = 1,610,510$ UZS

Growth is EXPONENTIAL — the interest earned each year is LARGER than the year before, because it compounds on a larger and larger base.

Feature	Simple Interest	Compound Interest
Formula	$\text{FV} = \text{PV} \times (1 + r \times n)$	$\text{FV} = \text{PV} \times (1 + r)^n$
Interest earned on	Original principal only	Principal + all accumulated interest
Growth pattern	Linear (straight line)	Exponential (curves upward)
1M at 10% for 10 years	2,000,000 UZS	2,593,742 UZS
1M at 10% for 20 years	3,000,000 UZS	6,727,500 UZS
1M at 10% for 30 years	4,000,000 UZS	17,449,402 UZS
1M at 10% for 40 years	5,000,000 UZS	45,259,256 UZS

■ Einstein's Eighth Wonder

"Compound interest is the eighth wonder of the world. He who understands it, earns it. He who doesn't, pays it." — Albert Einstein

After 40 years, compound interest produces 45M UZS while simple interest produces only 5M UZS — a difference of 40M UZS from the same 1M UZS investment. This is not magic. It is mathematics. And it works just as powerfully against you when you are the borrower paying compound interest on a loan.

The Rule of 72

The Rule of 72 is a quick mental shortcut that tells you how long it takes to **DOUBLE** your money at a given interest rate.

$$\text{Years to Double} \approx 72 \div \text{Interest Rate (\%)}$$

Interest Rate	Years to Double	Real Example
6%	$72 \div 6 = 12$ years	Government bonds
10%	$72 \div 10 = 7.2$ years	Stock market index
12%	$72 \div 12 = 6$ years	Bank deposit (UZS)
18%	$72 \div 18 = 4$ years	Good business investment
24%	$72 \div 24 = 3$ years	Excellent business / real estate
36%	$72 \div 36 = 2$ years	High-risk investment

— ■ WORKED EXAMPLE — WORKED EXAMPLE — The Cost of Waiting to Invest

Person A starts investing 2,000,000 UZS per year at age 22. Stops at age 32. Total invested: 20,000,000 UZS.

Person B starts investing 2,000,000 UZS per year at age 32. Continues to age 62. Total invested: 60,000,000 UZS.

Both earn 10% compound interest per year.

Result at age 62:

Person A (invested 20M, started early): approximately 338,000,000 UZS

Person B (invested 60M, started late): approximately 362,000,000 UZS

Person B invested **THREE TIMES** as much money but ends up with only slightly more — because they lost the most powerful years of compounding at the beginning.

If Person A had continued investing (not stopped at 32), they would have over 700,000,000 UZS.

The lesson: Time in the market is more powerful than the amount you invest. Start now.

2.4 Opportunity Cost of Capital

The **Opportunity Cost of Capital** is the return you give up by choosing one investment over the next best available alternative of similar risk. It is arguably the most important — and most overlooked — concept in personal finance.

Every financial decision has an opportunity cost. When you put money into one investment, you automatically forgo every other thing you could have done with that money. The opportunity cost is not the worst thing you gave up — it is the BEST thing you gave up.

Understanding Opportunity Cost: A Simple Example

You have 5,000,000 UZS. You are considering three options:

- Option A: Bank deposit at 14% per year (safe, guaranteed)
- Option B: Real estate investment, expected return 20% per year
- Option C: Start a small business, expected return 30% per year (but risky)

If you choose Option B (real estate), your opportunity cost is 14% — the safe guaranteed return you gave up from Option A. To justify choosing real estate over the bank, you must believe the extra risk of real estate is worth the extra 6% expected return (20% – 14%).

If you choose Option C (business), your opportunity cost is still 14% (the next best safe alternative). But now you are taking on significant additional risk for $30\% - 14\% = 16\%$ extra return. Is 16% enough compensation for the risk of losing everything? That is a judgment call — but opportunity cost gives you the framework to make it rationally.

The Opportunity Cost IS the Discount Rate

There is a direct connection between opportunity cost and Present Value. The discount rate you use in PV calculations IS the opportunity cost of capital. This is not a coincidence — it is the definition.

When you ask "what is a future cash flow worth today?", you are really asking: "How much would I need to invest today — at my opportunity cost rate — to replicate that future payment?" The present value IS the amount you'd invest today in your best alternative to produce the same future result.

Situation	Opportunity Cost Rate	Why
Investing in a safe government bond	12–14% (bank deposit rate)	You give up the guaranteed bank return
Investing in real estate	14–16% (safe alternatives)	Must beat your safest option
Company evaluating a project	Cost of capital (12–18%)	Must beat what shareholders could earn elsewhere

Paying off a 20% credit card	20% guaranteed	Paying off = earning 20% risk-free
Buying a car with savings	14%+ (investment return)	Cash spent = investment return foregone

The Hidden Costs of Everyday Decisions

Opportunity cost thinking transforms how you evaluate ordinary decisions. It reveals that "free" choices often have significant hidden costs.

- **Leaving cash in a low-interest account:** If your bank pays 5% but you could earn 14% in a deposit account, you are losing 9% per year on every sum you leave there. On 10,000,000 UZS, that is 900,000 UZS per year in foregone returns.
- **Keeping money at home:** Cash under a mattress earns 0%. With 12% inflation, you are losing 12% of purchasing power per year. This is not "safe" — it is a certain loss.
- **Buying an expensive item instead of investing:** A 5,000,000 UZS luxury purchase is not just a 5M cost. At 15% per year for 20 years, that money would have grown to 81,833,000 UZS. The true cost of the purchase is 81 million — the future value of what you gave up.

■ The Most Important Insight

The opportunity cost framework means there is NO SUCH THING as a "neutral" financial decision. Every choice you make with money — including choosing to do nothing — has a cost. The financially literate person always asks: "What am I giving up by making this choice?" and only proceeds if the benefits exceed the full opportunity cost.

2.5 Valuation of Assets

Valuation is the process of determining what an asset is **actually worth today**. It is one of the most important skills in all of finance — whether you are buying a stock, a property, a business, or a bond, you need to know: is the price being asked fair, too high, or too low?

The universal principle of valuation: Any asset is worth the present value of all the future cash flows it will generate for its owner. This single sentence contains the entire theory of valuation. Everything else is detail.

Value of Any Asset = PV of All Future Cash Flows

Net Present Value (NPV)

Net Present Value is the most important valuation tool in finance. NPV answers the question: **"If I invest money today and receive cash flows in the future, am I better off than if I had invested at my opportunity cost rate?"**

NPV is calculated by taking the present value of all future cash flows from an investment and subtracting the initial cost:

$$\text{NPV} = \sum [\text{CF}_t \div (1 + r)^t] - \text{Initial Investment}$$

Where CF_t is the cash flow in period t , r is the discount rate (opportunity cost of capital), and t runs from period 1 to the end of the investment.

NPV Result	What It Means	What You Should Do
NPV > 0 (Positive)	Investment returns MORE than your opportunity cost rate	✓ ACCEPT — creates value
NPV = 0 (Zero)	Investment returns EXACTLY your opportunity cost rate	Indifferent — break even
NPV < 0 (Negative)	Investment returns LESS than your opportunity cost rate	✗ REJECT — destroys value

— ■ WORKED EXAMPLE — WORKED EXAMPLE — NPV of a Business Investment

Scenario: You can buy a small shop for 50,000,000 UZS.

The shop will generate the following annual profits (after all costs):

Year 1: 8,000,000 UZS

Year 2: 10,000,000 UZS

Year 3: 12,000,000 UZS

Year 4: 12,000,000 UZS

Year 5: 15,000,000 UZS + sale of shop for 40,000,000 UZS = 55,000,000 UZS

Your discount rate (opportunity cost): 15% per year

PV of Year 1: $8,000,000 \div 1.15 = 6,956,522$

PV of Year 2: $10,000,000 \div 1.3225 = 7,561,437$

PV of Year 3: $12,000,000 \div 1.5209 = 7,890,257$

PV of Year 4: $12,000,000 \div 1.7490 = 6,861,963$

PV of Year 5: $55,000,000 \div 2.0114 = 27,344,481$

Total PV = $6,956,522 + 7,561,437 + 7,890,257 + 6,861,963 + 27,344,481 = 56,614,660$ UZS

NPV = $56,614,660 - 50,000,000 = +6,614,660$ UZS

Decision: NPV is POSITIVE — this investment creates 6.6M UZS of value above the opportunity cost. ACCEPT.

Valuing Stocks

A share of stock represents ownership in a company. Its value equals the present value of all future dividends plus the eventual sale price. In practice, analysts use several approaches:

- **Dividend Discount Model:** Value = Dividend $\div$ (Discount Rate – Dividend Growth Rate). Used for stable dividend-paying companies. Example: A company pays 5,000 UZS dividend per share, expected to grow 4% per year. If your discount rate is 14%, value = $5,000 \div (0.14 - 0.04) = 50,000$ UZS per share.
- **Price-to-Earnings (P/E) Ratio:** Stock Price $\div$ Earnings Per Share. A quick comparison tool. If the market average P/E is 15 and a company earns 10,000 UZS per share, a "fair" price is approximately 150,000 UZS. A stock trading below this is potentially undervalued.
- **Discounted Cash Flow (DCF):** Projects the company's free cash flows for 5–10 years, then discounts them back to the present. The most rigorous method, used by professional analysts.

Valuing Bonds

A bond is a loan you make to a company or government. In return, they promise to pay you fixed interest (coupon) payments at regular intervals, plus repay the principal at maturity. The value of a bond is simply the PV of all these future cash flows.

- **Coupon payments:** Fixed interest paid periodically (e.g., 8% of face value per year).
- **Face value (principal):** Repaid at the end of the bond's life (e.g., 1,000,000 UZS in 5 years).
- **Critical rule:** When market interest rates RISE, existing bond prices FALL — because the fixed future payments are now discounted at a higher rate, reducing their PV. When rates FALL, bond prices RISE.

Valuing Real Estate

Real estate generates value through rental income and property appreciation. The two main valuation approaches are:

- **Income Approach:** Value = Annual Net Rental Income ÷ Capitalization Rate. The "cap rate" is similar to a discount rate for property. Example: An apartment generates 24,000,000 UZS net per year. If the cap rate is 8%, value = $24,000,000 \div 0.08 = 300,000,000$ UZS.
- **Comparable Sales Approach:** Value based on recent prices of similar properties in the same area. Most commonly used in residential real estate.

— ■ WORKED EXAMPLE — WORKED EXAMPLE — Real Estate Valuation

Apartment in Tashkent: purchase price 200,000,000 UZS.

Annual rental income: 20,000,000 UZS. Annual expenses: 2,000,000 UZS.

Net income: 18,000,000 UZS per year.

Expected sale price after 10 years: 350,000,000 UZS.

Your required return: 12% per year.

PV of rental income (10 years at 12%):

Using annuity formula: $18,000,000 \times [(1 - (1.12)^{-10}) \div 0.12] = 18,000,000 \times 5.650 = 101,700,000$ UZS

PV of sale price: $350,000,000 \div (1.12)^{10} = 350,000,000 \div 3.1058 = 112,700,000$ UZS

Total Value = $101,700,000 + 112,700,000 = 214,400,000$ UZS

NPV = $214,400,000 - 200,000,000 = +14,400,000$ UZS

Decision: The apartment is worth MORE than it costs — a positive NPV investment. Buy it.

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Risk & Return

Risk and return are the two most fundamental concepts in investment decision-making, and they are inseparable. **In finance, higher expected return always comes with higher risk. Always. Without exception.** Anyone promising high returns with no risk is either misinformed or deliberately misleading you.

Understanding the risk-return relationship is not just academic. It is the framework that protects your savings from fraud, helps you build a rational investment portfolio, and allows you to make financial decisions without being paralyzed by fear or seduced by greed.

What Is Risk in Finance?

In everyday language, "risk" means the chance that something bad happens. In finance, risk has a more precise meaning: **risk is the uncertainty or variability of future returns.** An investment is "risky" not because it will definitely lose money, but because its future value is uncertain — it could be much higher or much lower than expected.

Asset Type	Expected Annual Return	Risk Level	What Could Go Wrong
Uzbekistan bank deposit	12–15%	Very Low	Bank failure (rare, often insured)
Government bonds (UZS)	16–20%	Low	Inflation erodes real return
Real estate (Tashkent)	15–25%	Medium	Market downturn, illiquidity, tenant issues
Index fund (S&P; 500)	8–12% (USD)	Medium-High	Market crashes (–40% in 2008)
Individual stocks	Variable	High	Company bankruptcy, fraud, competition
Bitcoin/Cryptocurrency	Very variable	Very High	–80% drops have occurred multiple times
Startup investment	0% or 10×+	Extreme	90% of startups fail within 5 years

Why Higher Return Demands Higher Risk

If a high-return investment were also completely safe, everyone would put all their money there. The flood of demand would drive up the price until the return fell to a level that reflected the risk. Markets are competitive: **any genuine free lunch gets arbitrated away immediately.** High returns exist only because investors demand extra compensation for bearing extra uncertainty.

This is why a 30% return from a startup sounds attractive — until you remember that most startups fail completely. The high expected return is compensation for the very real probability of losing everything.

Diversification: The One Free Lunch

While the risk-return tradeoff cannot be eliminated entirely, it can be managed through **diversification** — spreading investments across multiple assets that do not move together perfectly.

When one investment falls, another may rise or hold steady. The overall portfolio becomes more stable than any individual holding, without necessarily sacrificing expected return. This is why diversification is called "the only free lunch in finance."

- **Within asset classes:** Do not put all money in one stock — own 15–20+ across different industries
- **Across asset classes:** Mix deposits, bonds, real estate, and stocks
- **Across geographies:** Invest in Uzbekistan AND international markets (USD assets protect against UZS devaluation)
- **Across time:** Invest regularly (monthly) rather than all at once — reduces timing risk

— ■ WORKED EXAMPLE — WORKED EXAMPLE — The Cost of Concentration

Scenario A — Concentrated: You invest all 20,000,000 UZS in one company stock.

The company goes bankrupt. You lose everything. Portfolio value: 0.

Scenario B — Diversified: You invest 1,000,000 UZS each in 20 different companies.

Three companies go bankrupt (a very bad year). You lose 3,000,000.

Portfolio value: 17,000,000 UZS — you are still in the game.

Both portfolios started identically. The only difference was concentration vs. diversification. Diversification did not prevent losses — but it prevented catastrophic, unrecoverable loss.

This is the purpose of diversification: not to maximize returns, but to survive bad outcomes.

Building a Simple Investment Pyramid

For a beginner investor in Uzbekistan, a simple pyramid approach helps structure your portfolio by risk level:

- **Foundation (50–60%):** Safe, liquid assets — bank deposits, government bonds. These are your emergency fund and low-risk base. Never put this at risk.
- **Middle layer (25–30%):** Medium-risk investments — real estate, established businesses, index funds. Long-term growth with manageable volatility.
- **Top layer (10–20%):** Higher-risk investments — individual stocks, startup investments. Only money you can afford to lose entirely. High potential upside.

■ Key Takeaway for 2.6

Never invest money you cannot afford to lose in high-risk assets. Never chase high returns without understanding what risk you are taking on. The goal of investing is not to maximize return — it is to maximize return FOR A GIVEN LEVEL OF RISK YOU CAN COMFORTABLY BEAR. Build your foundation first. Grow from there. Be patient. Compound interest and diversification are your most powerful tools.

Key Terms Glossary

All Important Definitions from Unit 2

Time Value of Money	The principle that money available today is worth more than the same amount in the future, because today's money can be invested to earn returns, purchasing power is higher today, and the future is uncertain.
Present Value (PV)	The current worth of a future sum of money, discounted at an appropriate rate. Answers: "What is this future payment worth to me today?" Formula: $PV = FV \div (1 + r)^n$.
Future Value (FV)	The value that a sum of money invested today will grow to at a specified future date, given a certain rate of return. Formula: $FV = PV \times (1 + r)^n$.
Discount Rate	The interest rate used to convert future cash flows into their present value. Represents the opportunity cost of capital — the return you could earn on an alternative investment of similar risk.
Compound Interest	Interest calculated on both the original principal AND all previously accumulated interest. Produces exponential growth over time. The most powerful force in personal wealth building.
Simple Interest	Interest calculated only on the original principal, not on accumulated interest. Produces linear (straight-line) growth. Less powerful than compound interest over time.
Opportunity Cost of Capital	The return you give up by choosing one investment over the next best alternative of similar risk. This IS the discount rate used in present value calculations.
Net Present Value (NPV)	The sum of the present values of all future cash flows from an investment, minus the initial cost. $NPV > 0$ means the investment creates value; $NPV < 0$ means it destroys value.
Valuation	The process of determining the current fair value of an asset or company. Based on the universal principle that any asset is worth the present value of all the cash flows it will generate.
Intrinsic Value	The true economic value of an asset based on fundamental analysis — the present value of all expected future cash flows. May differ significantly from market price.
Discount Factor	The multiplier $(1 + r)^{-n}$ used to convert a future cash flow to present value. Always less than 1 (for positive r), reflecting that future money is worth less today.
Annuity	A series of equal cash flows paid or received at regular intervals over a specified period. Common examples: mortgage payments, bond coupons, rental income.

Diversification	The strategy of spreading investments across multiple assets, industries, and geographies to reduce risk without necessarily reducing expected return. The "only free lunch" in finance.
Risk-Return Tradeoff	The fundamental principle that higher potential returns always come with higher risk. Investors require higher expected returns to compensate for bearing greater uncertainty.
Capitalization Rate (Cap Rate)	Used in real estate: $\text{Cap Rate} = \text{Net Annual Income} \div \text{Property Value}$. Used to quickly value income-producing properties and compare investment opportunities.
Rule of 72	A mental shortcut: $\text{Years to double your money} \approx 72 \div \text{Annual Interest Rate (\%)}$. Quick way to estimate the effect of compound growth without a calculator.

Practice Quiz

12 Questions Covering Sections 2.1 through 2.6

These questions test your understanding of all six sub-sections. Select the best answer for each question. The answer key with brief explanations is at the end of this section.

Q1 [Section 2.1]

You will receive 5,000,000 UZS in 3 years. A friend offers to give you 4,000,000 UZS today instead. The bank deposit rate is 10% per year. Which should you choose and why?

- A) Take the 4,000,000 today because cash in hand is always better
- B) Take the 4,000,000 today because PV of 5M in 3 years at 10% is only 3,756,574 — less than 4M
- C) Wait for 5,000,000 because it is a larger number
- D) It makes no difference — money has no time value

Q2 [Section 2.1]

Which of the following CORRECTLY explains why interest exists on loans?

- A) Banks invented interest to make profit
- B) Governments require interest to control inflation
- C) Interest compensates lenders for giving up investment opportunities, inflation risk, and uncertainty
- D) Interest is only charged when there is a risk of default

Q3 [Section 2.2]

Using $PV = FV \div (1 + r)^n$, what is the present value of 12,000,000 UZS to be received in 2 years, at a discount rate of 15%?

- A) 10,400,000 UZS
- B) 9,075,145 UZS
- C) 10,000,000 UZS
- D) 11,000,000 UZS

Q4 [Section 2.2]

Two investments both promise to pay 20,000,000 UZS in 5 years. Investment A has a discount rate of 10%. Investment B has a discount rate of 20%. Which has the higher present value?

- A) Investment B because 20% is a higher number
- B) They are equal because they both pay 20,000,000
- C) Investment A because lower discount rate means higher PV
- D) Cannot determine without more information

Q5 [Section 2.3]

You invest 3,000,000 UZS at 12% compound interest per year for 4 years. Approximately what is the future value?

- A) 4,440,000 UZS
- B) 4,716,349 UZS

- C) 4,000,000 UZS
- D) 5,200,000 UZS

Q6 [Section 2.3]

A bank offers 12% simple interest or 10% compound interest per year for 5 years. On 1,000,000 UZS, which is better?

- A) 12% simple interest (1,600,000 UZS) because the rate is higher
- B) 10% compound interest (1,610,510 UZS) because compounding produces more
- C) They are exactly equal over 5 years
- D) Simple interest is always better than compound interest

Q7 [Section 2.4]

You have 8,000,000 UZS. You invest it in stocks expecting 18% per year. The bank offers 14% guaranteed. One year later the stocks returned only 11%. What was your opportunity cost?

- A) 11% — the actual return you earned
- B) 18% — the expected return
- C) 14% — the bank deposit you gave up
- D) 7% — the difference between expected (18%) and actual (11%)

Q8 [Section 2.4]

An investment returns 16% per year. Your opportunity cost of capital is 14%. Should you invest?

- A) No — 16% is not high enough
- B) Yes — the return exceeds the opportunity cost, so the investment creates value
- C) Only if there is no risk at all
- D) Only if the investment lasts at least 10 years

Q9 [Section 2.5]

You invest 30,000,000 UZS in a project. The present value of all future cash flows is 26,000,000 UZS. What is the NPV and what should you do?

- A) NPV = +26,000,000 — Accept
- B) NPV = -30,000,000 — Reject
- C) NPV = -4,000,000 — Reject, this investment destroys value
- D) NPV = +4,000,000 — Accept

Q10 [Section 2.5]

A bond pays 10,000,000 UZS per year for 3 years plus 100,000,000 UZS at maturity. Market interest rates suddenly rise sharply. What happens to this bond's price?

- A) The bond price rises because higher rates mean more income
- B) The bond price falls because the fixed payments are now discounted at a higher rate, reducing PV
- C) Nothing changes — bonds are not affected by interest rates
- D) The bond price doubles

Q11 [Section 2.6]

Which of the following statements about risk and return is CORRECT?

- A) It is possible to find genuinely high-return, zero-risk investments if you look hard enough
- B) Lower risk always means higher returns because safe companies are well-managed
- C) Higher expected return always comes with higher risk — there are no exceptions
- D) Risk only matters for large investors, not for individuals

Q12 [Section 2.6]

An investor puts 50% in bank deposits, 30% in real estate, and 20% in index funds. This is BEST described as an example of:

- A) Speculation
- B) Compound interest
- C) A diversified portfolio structured as an investment pyramid
- D) Opportunity cost calculation

Answer Key with Explanations

Q1: B	$PV = 5,000,000 \div (1.10)^3 = 5,000,000 \div 1.331 = 3,756,574$ UZS. Since $4M > 3.75M$, take the 4M today.
Q2: C	Interest compensates lenders for all three: foregone investment returns, inflation erosion, and default risk.
Q3: B	$12,000,000 \div (1.15)^2 = 12,000,000 \div 1.3225 = 9,075,145$ UZS.
Q4: C	Higher discount rate = lower PV. Investment A (10%) has higher PV than Investment B (20%) for the same future amount.
Q5: B	$3,000,000 \times (1.12)^5 = 3,000,000 \times 1.57352 = 4,720,565$ UZS $\approx 4,716,349$ UZS.
Q6: B	Simple: $1,000,000 \times (1 + 0.12 \times 5) = 1,600,000$. Compound: $1,000,000 \times (1.10)^5 = 1,610,510$. Compound wins.
Q7: C	Opportunity cost = what you gave up = 14% bank deposit rate. Actual vs. expected return is a different concept.
Q8: B	Return (16%) > Opportunity cost (14%). The investment creates value above what you could earn elsewhere. Accept.
Q9: C	$NPV = 26,000,000 - 30,000,000 = -4,000,000$. Negative NPV means the investment destroys value. Reject.
Q10: B	When rates rise, discount rate increases, so PV of all fixed future bond payments falls. Bond price falls.
Q11: C	The risk-return tradeoff is a fundamental law of markets. High return always comes with high risk.
Q12: C	Mixing deposits, real estate, and index funds across different risk levels is a classic diversified pyramid portfolio.

■ Score Guide

12/12: Perfect — you have fully mastered Unit 2. You are ready for Unit 3.

10–11: Excellent — minor gaps. Review the questions you missed and re-read that sub-section.

8–9: Good — re-read sections 2.3 and 2.5 carefully, focusing on the worked examples.

6–7: Fair — go back through all six sub-sections with the glossary open.

Below 6: Re-study the full guide from the beginning. Focus on the formulas and examples.